MOCK Second Partial Exam

Question 1 (7.5 points)

In country Alpha there are only one-year and two-year bonds. Current and expected future inflation are equal to zero, so nominal and real interest rates are the same. In period $t+1$ ("the future") a standard IS-LM model describes the functioning of the economy: consumption depends only on disposable income at time $t+1$, while investment depends on the values that interest rate and income will assume at time $t+1$. As for the current period (t, "the present"), consumption depends positively on disposable income at time t and on the expected disposable income at time $t+1$, $C_t=C(Y_t-T_t, Y^e_{t+1}-T^e_{t+1})$, and investment is exogenous. *G and T* are exogenous as usual in both periods. Furthermore, at time *t and t+1*, in the initial equilibrium position both income and the interest rate take on positive values. Suppose now that, at time *t*, the government announces that at time *t+1* taxation (T) will be reduced and at the same time the central bank announces that it will implement at time t+1 a monetary policy aimed at keeping constant the level of output at t+1. To achieve its goal, what kind of monetary policy should the central bank implement at t+1? Explain the effects of the two announcements on Alpha's current (time t) income and interest rate equilibrium levels.

Question 2 (7.5 points)

"Assume that in country Beta some consumers are subject to "liquidity constraints" and they cannot get credit: if the government of Beta announces that at time *t+1* taxation (T) will be reduced, demand and output in Beta will increase in the future (t+1) more than in a country without liquidity constraints." True or False? Explain.

Question 3 (7.5 points)

a) Consider country Gamma, open to foreign trade in goods, services and financial assets and operating under a flexible exchange rate regime, which can be described with the model "IS−LM - interest parity relation". Assume that there is an increase in the autonomous component of investment and explain how Gamma's net exports will change. Suppose that later on Gamma's economic policy authorities agree to adopt measures designed to bring back net exports towards its initial level, while keeping the level of income to the new level. Which economic policies should be implemented in order to achieve both goals? Explain.

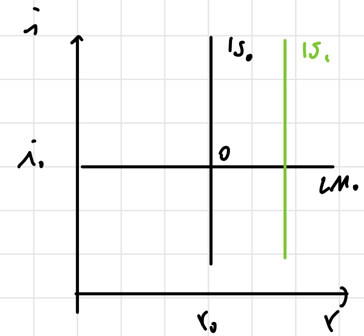
Question 4 (7.5 points)

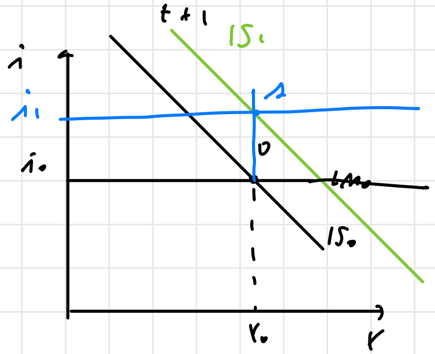
Explain whether the following statements are ***true or false****. Motivate - briefly but rigorously - your answer by referring to the relevant theory. Answers inadequately motivated or without reference to the economic intuition will receive zero points.*

a) "If $B_{t-1}/Y_{t-1} > 0$, $r < g$, and $(G_t - T_t) / Y_t < 0$, then necessarily $B_t/Y_t > B_{t-1}/Y_{t-1}$."

b) "If investment is not sensitive to the interest rate, the announcement (unexpected by individuals) that, from time *t* onwards, monetary policy will become permanently more expansive does not make the price of shares at time *t* change."

1.





start at $t+1$: $T_{t+1}\downarrow$ increase in disposable income

$\Rightarrow C\uparrow,\ Z\uparrow,\ Y\uparrow$ IS to the right

$\Rightarrow$ CB: $i\uparrow$ contract. mon. policy

At time t: $r^e = i^e\uparrow$

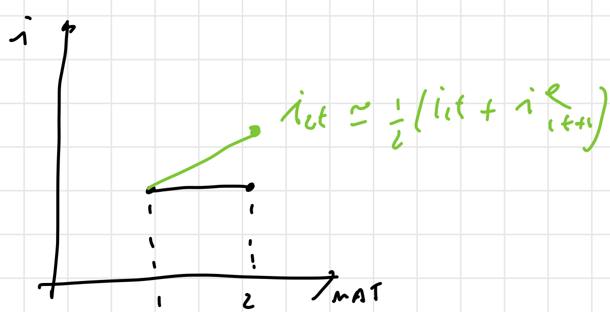
$T^e_{t+1}\downarrow \Rightarrow C\uparrow,\ Z\uparrow, Y\uparrow$

I exogenous!!!

$\Rightarrow \Delta I = 0$

$i =$ $Y\uparrow$

$$\boxed{Q_t\downarrow \left\{ \begin{array}{l} \underset{=}{D^e}\ (\underset{=}{r^e}) \\ \underset{-}{r} = \\ \underset{-}{r}^e\uparrow \end{array} \right.}$$



Question 2 (7.5 points)

"Assume that in country Beta some consumers are subject to "liquidity constraints" and they cannot get credit: if the government of Beta announces that at time *t+1* taxation (T) will be reduced, demand and output in Beta will increase in the future (t+1) more than in a country without liquidity constraints." True or False? Explain.

1

consumpt. will ↑ more in the future

bc w/ L.C. people will smooth more

consumption

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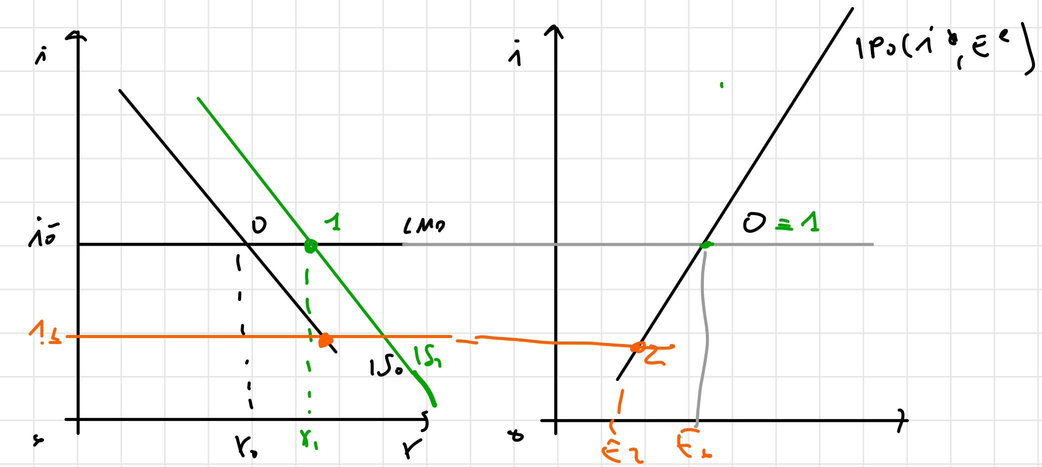
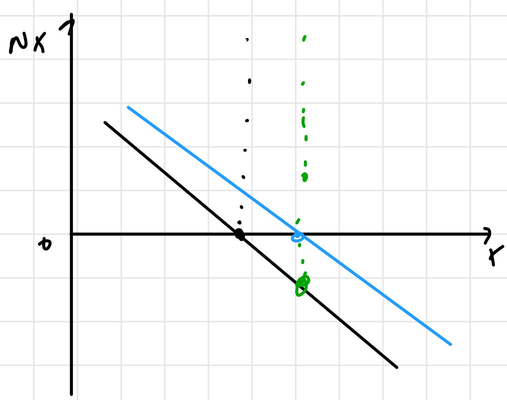




$\bar{I}\uparrow \Rightarrow I\uparrow, Z\uparrow, Y\uparrow$

$Y_1 > Y_0$

$i =$

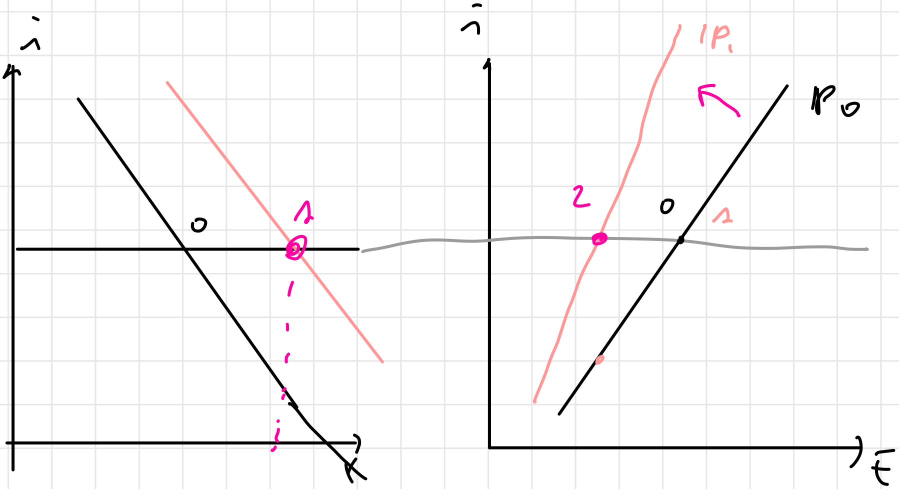
$E =$ competit. is the same

$NX\downarrow \begin{cases} E = \downarrow \\ Y^* = \\ Y\uparrow \ (IM\uparrow) \end{cases}$ If (I want to bring back NX to 0)

1) exp. mon. policy in order to decrease i
$i_2 < i_0$
$\Rightarrow E_2 < E_0$

2) fiscal contraction ($G\downarrow$ / $T\uparrow$)

$Y = Y_1$
$NX_2 > NX_1$



$\bar{E}^e_1 < \bar{E}^e_0$

$$NX\uparrow \begin{cases} E^e \downarrow \\ E = \\ K = \end{cases}$$


TROVA DIO!

Question 4 (7.5 points)

Explain whether the following statements are ***true or false****. Motivate - briefly but rigorously - your answer by referring to the relevant theory. Answers inadequately motivated or without reference to the economic intuition will receive zero points.*

$2 + r - \rho$ — $\rho > r$, < 2 — $r > \rho$, > 2

surplus

a) "If $B_{t-1}/Y_{t-1} > 0$, $r < g$, and $(G_t - T_t) / Y_t < 0$, then necessarily $B_t/Y_t > B_{t-1}/Y_{t-1}$."

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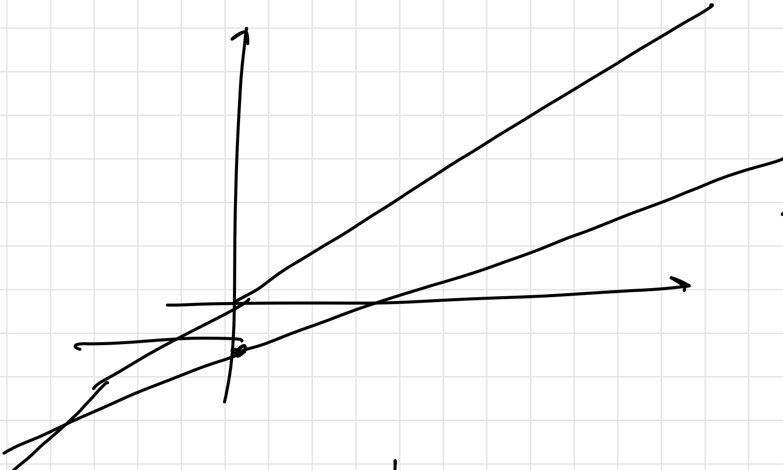
(a)



B

$$\bar{b} = \frac{\frac{G-T}{Y}}{\rho - r} \quad \begin{matrix} (-) < 0 \\ (+) \end{matrix}$$

since $b_0 > \bar{b} \Rightarrow \frac{B}{Y} \downarrow$ over time

$$\left.\begin{matrix} r^e \downarrow \\ r \downarrow \\ Q^e = (Y^e\ :) \end{matrix}\right\} Q_t \uparrow$$